

Market Making with \$1,000,000 — Can We Do It?

Feasibility & go-to-market brief · 21 Jun 2026 · Maximus + 3 CC research streams (data audit · equity economics · options MM)

Short answer: Yes, market making is possible to start with \$1MM — but **not** the kind most people picture (quoting SPY all day for rebates). That arena is owned by colocated HFT and is a money-loser for us. The version that *works* at our size lives in **wide-spread instruments**: options and crypto. And the single highest-ROI move isn't building a standalone desk at all — it's **turning our existing v8a edge into a liquidity-providing strategy**, which we can backtest *today*.

✗ NOT VIABLE

Classic equity/ETF market making (SPY, QQQ) at \$1MM, non-colocated. The tick is \$0.01 on a ~\$600 ETF (0.0017% spread). After adverse selection a realistic round trip nets $\approx -\$0.0067/\text{share}$ at a ~65% toxic-fill rate. You'd need **~\$300MM/day turnover ($\approx 300\times$ capital)** just to *attempt* \$1k/day — and you'd lose money doing it. The rebate and the queue belong to Citadel/Virtu. Latency is the binding constraint here, and we can't win it.

✓ VIABLE — START HERE

Liquidity provision on our own v8a flow (options), + a ring-fenced options-MM pilot. We already have a validated VRP/credit-spread edge on SPY/QQQ/XLF/XLI/GLD/SLV. Today we *pay* the wide option spread on every entry and exit. By posting **inside the NBBO** instead of crossing it, we capture part of that spread — a multi-percent P&L uplift, near-zero added tail risk, and **backtestable now on data we own**.

~ POSSIBLE, EARNED

Crypto spot MM (mid-cap pairs) and standalone options MM. Real spreads (30–50bp crypto; 1–5% on illiquid option strikes), no PFOF, no

registration barrier, 24/7 at-bats for crypto. But the edge is *conditional on execution quality* and carries genuine tail risk. Realistic well-run book: ~\$300-\$1,500/day on \$1MM in normal conditions, high variance, with real drawdowns. This is a quant operation, not hot-keys.

1. Is \$1MM enough? Yes — capital is not the binding constraint

\$1MM is plenty to *start*. Market making isn't gated by capital; it's gated by three other things, in order:

Spread width

WHERE THE EDGE LIVES

Adverse selection

WHAT EATS THE EDGE

Execution / latency

BINDING ONLY IN LIQUID NAMES

Modeling + ops

OUR ACTUAL CAPACITY LIMIT

The implication: at \$1MM you don't scale by trading *more size in one liquid name* — you scale by running **many small, risk-capped books** (15–40 instruments) where spreads are wide enough that adverse selection doesn't erase the edge. Capital per book is tiny; the constraint is how many books our models and risk system can run well.

2. The brutal math — why liquid equity MM fails for us

Component	Liquid ETF (SPY)	Why it matters
Quoted half-spread	~0.083 bp (\$0.005)	Tick-constrained at \$0.01; almost nothing to capture
Maker rebate (if DMA)	+\$0.0020–0.0030/sh	Needs cost-plus broker (IBKR Pro/Lime/Cobra); PFOF brokers keep it
Adverse selection	–(dominates)	~65% of passive fills are toxic; you're picked off by faster quotes
Net / round trip	≈ –\$0.0067/sh	Negative <i>after</i> rebate at realistic toxicity

Turnover for ~\$1k/day	~\$300MM/day (~300x)	Physically impossible for a non-colo \$1MM book
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Frameworks: Avellaneda-Stoikov (optimal quoting), Glosten-Milgrom & Ho-Stoll (adverse selection / inventory). Conclusion is robust: in tick-constrained liquid names, **latency is the moat and we don't have it**. Step into wider spreads and the binding constraint shifts to *modeling and risk* — which a disciplined small operator can win.

3. Can we even backtest it? Honest data audit

We inspected every relevant file on disk. Here's the truth:

Asset for MM	Data we have	Verdict
Equity / ETF MM	None. Zero equity quote or trade data on disk (only a reference <code>underlying_price</code> scalar)	Can't backtest
Options MM (standalone desk)	CBOE SPX/SPY/QQQ with real per-strike bid/ask + greeks , but only hourly (8 bars/day, starts 10:00)	Crude upper-bound only
Options trade history	<code>options_cache.db</code> : 6.3M daily + 1.6M 5-min trade bars — no bid/ask	Reference only
v8a execution upgrade (post inside NBBO)	EOD option quotes we already use for v8a backtests	Backtestable today

The single biggest gap: we have *no event-level (tick) NBBO + order-book depth*, so we can't build a credible **fill/queue model** — and in market making, the fill model *is* the strategy. Hourly quotes can sketch an optimistic P&L; they cannot prove a real quoting business works.

Also noted: the existing `reports/market_making_sim.html` is a 100% synthetic Avellaneda-Stoikov toy that has never touched a real price. It tells us nothing about live edge — by design it lives in `archive/` (it violates our Rule Zero "no synthetic data").

4. Options MM — real edge, but mind the wrapper

Wide option spreads are the natural home for a small market maker. But there's a sharp insight from the research: **an options MM book run net-short is just our v8a VRP edge made continuous, naked, and uncapped** — i.e. strictly *worse* risk for marginally more return. The same "VIX→80 = 100% breach" tail that v8a survives *because of defined-risk wings* becomes an open-ended blow-up for a naked MM desk.

- **P&L = spread capture + theta – negative-gamma hedging losses – vega MTM – adverse selection.** The greeks *are* the business; you must delta-hedge continuously.
- **Honest expectation:** Sharpe ~1.5–3.0 (not v8a's 6.4), base case ~8–20%/yr on \$1MM, with a fat left tail.
- **You can't beat wholesalers for the good flow:** at \$1MM you're a prop/retail account, not a registered MM — no rebates, no membership; the clean retail flow is internalized via PFOF. Your resting quote eats the adversely-selected residual.
- **Niche that fits us:** NOT SPX 0DTE. Mid-liquidity ETF options (XLF, XLI, GLD, SLV, IWM) and wide-spread single names — exactly where we already have VRP signal and data.

5. Recommendation — the path that actually starts

Don't build a standalone MM desk first. Convert the edge we already have, prove adverse selection live with tiny capital, then decide on scale.

- 1 **v8a execution upgrade (now, ~zero added risk).** On every v8a entry/exit, post *inside the NBBO* instead of crossing the spread. We currently pay multi-percent in spread we could be capturing.

Backtestable today on EOD data

- 2 **Ring-fenced inside-the-spread pilot ($\leq$ \$100k).** On XLF/XLI/SLV — where we have signal — post passive 2-sided quotes purely to **measure live adverse selection / markouts**. This is the number every model needs and no backtest can give us honestly. Cap risk hard.
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- 3 **Buy the data before believing any MM backtest.** For a real options-MM sim we need 1-min / OPRA TAQ quotes on the Tier-1 ETF underliers. Until then, treat hourly-CBOE results as an *upper bound*, not evidence.
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- 4 **Optional parallel track — crypto spot MM spike.** Mid-cap pairs (30–50bp spreads, no PFOF, 24/7). Small allocation, automated, inventory-skew quoting. Best *accessible* standalone MM, but earn it: at low fee tiers the maker fee can exceed a tight spread — pair selection + volume tiers decide profitability.
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- 5 **Scale by count, not size.** If the pilot's measured markouts are favorable, deploy across 15–40 small risk-capped books rather than concentrating. Capacity = modeling/ops bandwidth, not capital.
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6. Bottom line

Is \$1MM enough to start? Yes. Capital isn't the constraint — spread width, adverse selection, and our modeling/ops bandwidth are.

Can we "crush" liquid equity MM? No. That's an HFT/latency game we lose. Don't play it.

Where we can win: wide-spread liquidity provision — first by upgrading v8a's execution (free money we currently leave on the table), then a measured options/crypto MM pilot. Expect a well-run \$1MM MM book to make **~\$300–\$1,500/day** in normal conditions — real, but variance-heavy and tail-exposed, and *conditional on execution quality*.

First move I'd green-light: backtest the v8a inside-the-NBBO execution upgrade (Step 1). It's the highest ROI, lowest risk, and we can run it this week. Say the word and I'll put CC on it.

Sources (full detail in repo): [research/MM_DATA_INFRA_AUDIT.md](#) · [research/MM_EQUITY_FEASIBILITY.md](#) · [research/MM_OPTIONS_FEASIBILITY.md](#). All data figures read from live files on 21 Jun 2026; no values estimated or fabricated. Concepts cited: Avellaneda-Stoikov (2008), Glosten-Milgrom (1985), Ho-Stoll (1981), Garleanu-Pedersen-Poteshman.